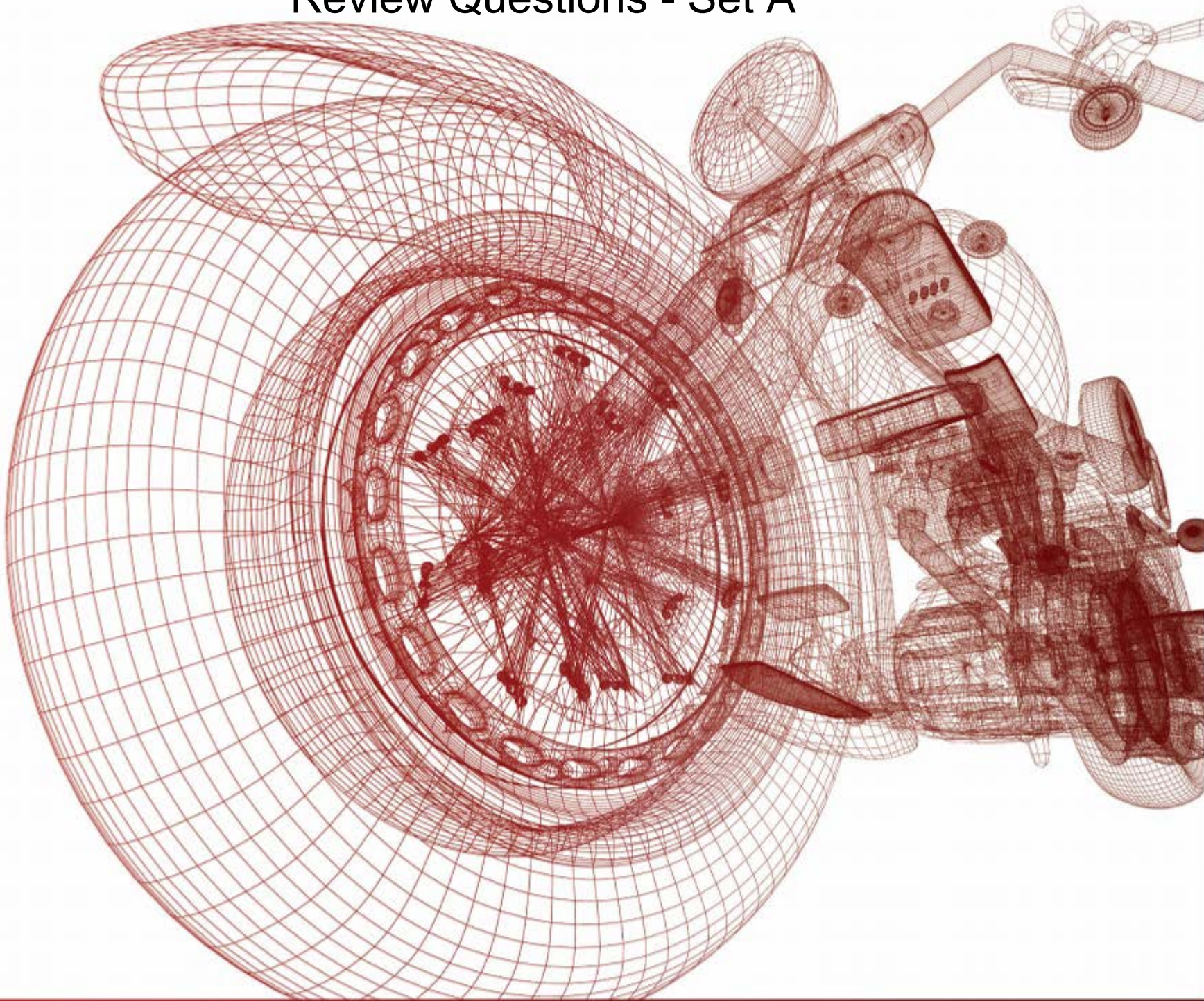




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Inventor 2022 - Introduction to Solid Modeling Review Questions - Set A



Autodesk[®] Inventor[®] 2022
Introduction to Solid Modeling
Learning Guide (Mixed Units) - 1st Edition

 **AUTODESK**
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Chapter Review Questions

1. Match the numbers shown in Figure 1–52 with the interface components listed below.

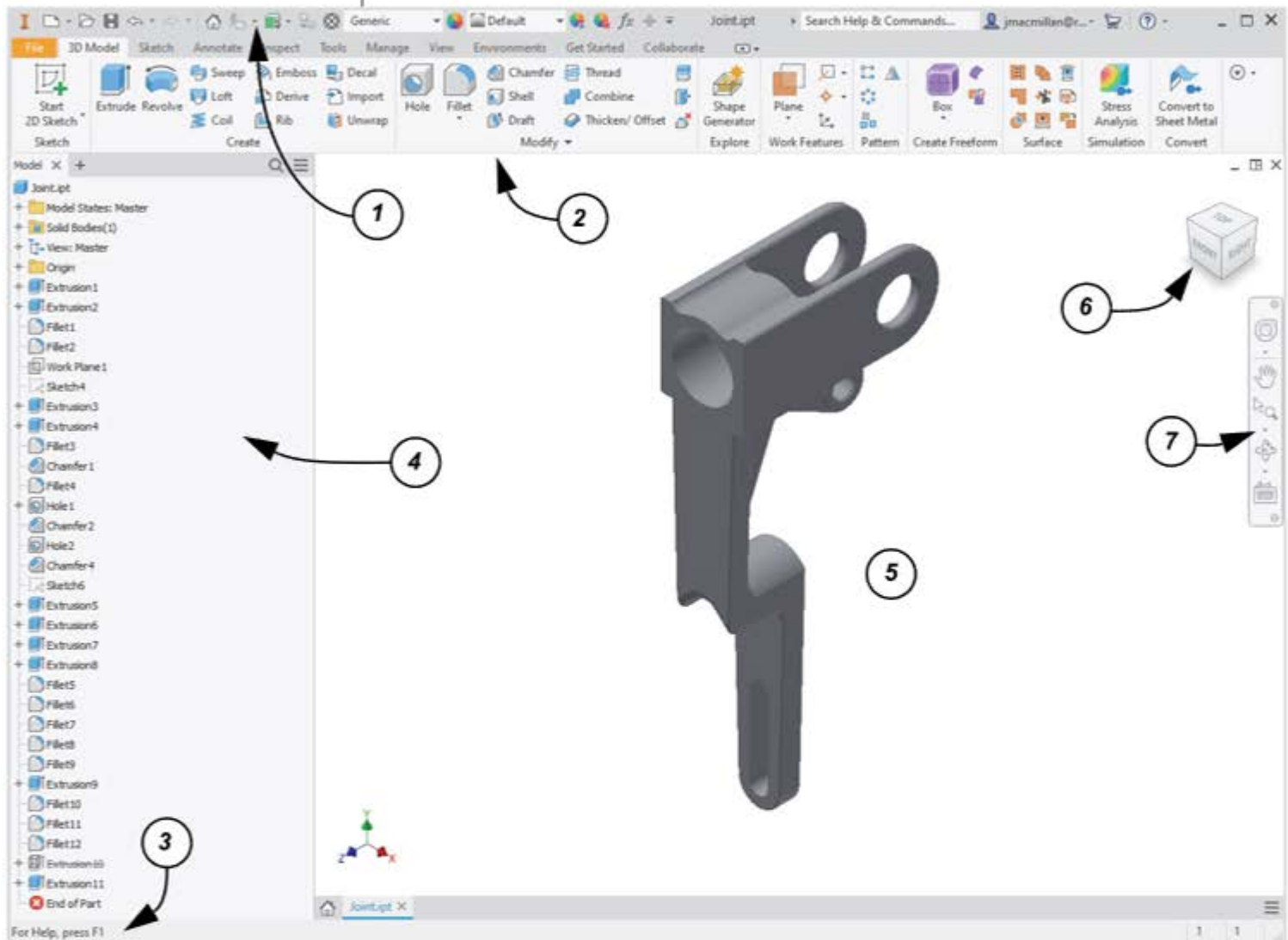


Figure 1–52

- | | |
|-------------------------|-------|
| a. Navigation Bar | _____ |
| b. Model Browser | _____ |
| c. Status Bar | _____ |
| d. Quick Access Toolbar | _____ |
| e. Ribbon | _____ |
| f. Graphics Window | _____ |
| g. ViewCube | _____ |

2. After editing the dimensions of a part model, you must open all drawings referencing that part to make the same dimension changes.
 - a. True
 - b. False
3. Which of the following are valid filename extensions for Autodesk Inventor files? (Select all that apply.)
 - a. .IPT
 - b. .IDW
 - c. .IAM
 - d. .INV
4. Clicking  enables you to automatically reorient the display of the model to its top view.
 - a. True
 - b. False
5. Which mouse button do you click and hold to pan the model without activating the **Pan** command?
 - a. Left
 - b. Middle
 - c. Right
6. Which combination of items do you select to quickly orient a model face parallel to the screen without spinning? (Select all that apply.)
 - a. A surface and .
 - b. A planar surface and .
 - c. A surface and .
 - d. A planar surface and .
 - e. A face on the ViewCube.
7. You can open multiple windows in one session.
 - a. True
 - b. False

8. What is the purpose of the menu shown in Figure 1–53?



Figure 1–53

- a. To switch back to a previous view or ahead to a current view.
- b. To cycle through different objects for selection.
- c. To pan the view left or right.
- d. To zoom in or out on the selected object.

Chapter Review Questions

- What template would you use to create a new standard Part file?
 - Standard.ipn**
 - Standard.idw**
 - Standard.ipt**
 - Standard.iam**
- Which of the following can be defined before sketching geometry? (Select all that apply.)
 - Sketch Plane
 - Sketch References
 - Constraints
 - Dimensions
- An Origin Plane cannot be selected as a sketch plane.
 - True
 - False
- Match the command in the left column to its interface icon in the right column.

a. Dimension



Answer

b. Automatic Dimensions and Constraints



c. Extrude



d. Show Constraints



e. Project Geometry



5. The **Automatic Dimensions and Constraints** option controls whether all of the dimensions and/or constraints required to constrain a sketch are automatically added. Additionally, it controls how the dimensions are positioned in the sketch.
 - a. True
 - b. False
6. The sketched entities shown on the left in Figure 2–55 are fully constrained. Which of the following statements best describes the (113.91) dimension value, as shown on the right in Figure 2–55?

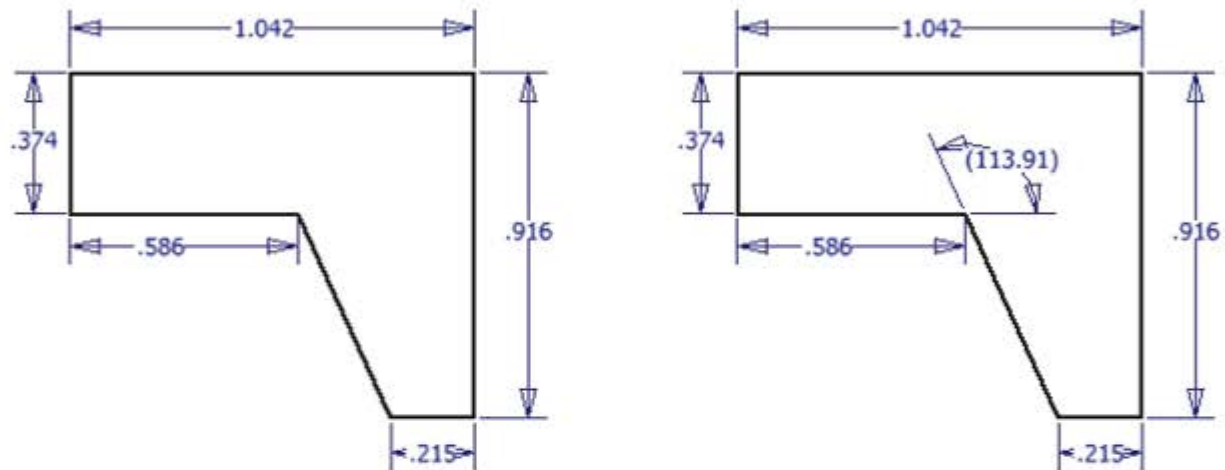


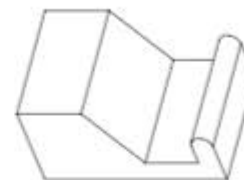
Figure 2–55

- a. The (113.91) dimension is a Driven Angular Dimension and cannot be modified to change the size of the sketch.
- b. The (113.91) dimension is a General Angular Dimension and cannot be modified to change the size of the sketch.
- c. The (113.91) dimension is a Driven Angular Dimension and can be modified to change the size of the sketch.
- d. The (113.91) dimension is a General Angular Dimension and can be modified to change the size of the sketch.

7. Match the feature form name in the left column with the geometry that would be created using this type of feature.

Answer

a. Revolve



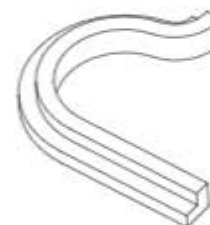
b. Sweep



c. Loft



d. Extrude



8. Which option do you use to open the feature's Properties panel, which would enable you to make a change to the feature (such as flipping the extrusion direction)?

- a. **Edit Sketch**
- b. **Show Dimensions**
- c. **Edit Feature**
- d. **Properties**

9. What option enables you to change the sketch plane of a feature?

- a. **Show Dimension**
- b. **Edit Feature**
- c. **Edit Sketch**
- d. **Redefine**

10. What option enables you to make changes to a sketch so that you could delete entities?
- a. **Show Dimension**
 - b. **Edit Feature**
 - c. **Edit Sketch**
 - d. **Redefine**

Chapter Review Questions

1. Construction entities can be used directly to create solid geometry.
 - a. True
 - b. False
2. Which of the following statements describes Trimming sketch entities?
 - a. Continues an entity to meet the next entity in its path or to close an open sketch.
 - b. Removes the segment of the entity to the nearest intersection in each direction from the point selected.
 - c. Duplicates a source entity to one or more locations in the sketch or to another sketch.
 - d. Provides a specific tool for stretching sketched entities versus simply selecting and dragging them in the sketch.
3. Which of the following statements describes Extending sketch entities?
 - a. Continues an entity to meet the next entity in its path or to close an open sketch.
 - b. Enables you to relocate original 2D sketch geometry.
 - c. Duplicates a source entity to one or more locations in the sketch or to another sketch.
 - d. Provides a specific tool for stretching sketched entities versus simply selecting and dragging them in the sketch.
4. How was the profile, shown in Figure 3–71, created in sketch mode?

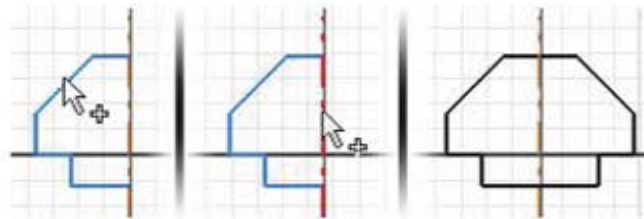


Figure 3–71

- a. **Mirror**
- b. **Revolve**
- c. **Extend**
- d. **Trim**

5. Which of the following can be made equal using  (Equal Constraint)? (Select all that apply.)
- a. Angles between lines
 - b. Line Lengths
 - c. Arc Radii
 - d. Circle Diameters
6. Which constraints enable you to create two circles with the same center? (Select all that apply.)
- a. Concentric
 - b. Colinear
 - c. Coincident
 - d. Tangent
7. How can you control the visibility of constraint symbols in the graphics window display? (Select all that apply.)
- a. To show or hide all of the constraints on a sketch, toggle  (Show All Constraints) and  (Hide All Constraints) in the Status Bar.
 - b. Select the entity to which it is associated.
 - c. Use <F8> to show all of the constraints and <F9> to hide them.
 - d. Select the symbol and press <Delete>.
8. What is the purpose of the Fix constraint, shown in Figure 3–72?

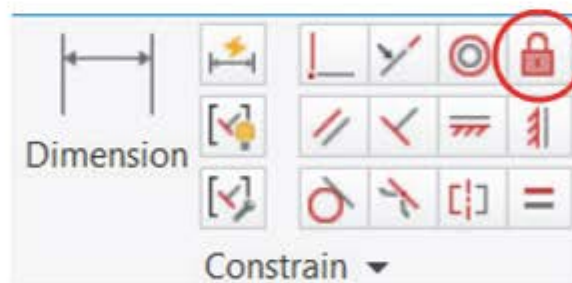


Figure 3–72

- a. To position one point exactly on another.
- b. To edit an existing constraint.
- c. To resolve conflicts among other constraints.
- d. To fix a point relative to the default coordinate system of the sketch.

9. All constraints must be manually assigned to the sketched geometry.
- True
 - False
10. Which of the following sets of constraints is used to manipulate the geometry on the left to that on the right in Figure 3–73?

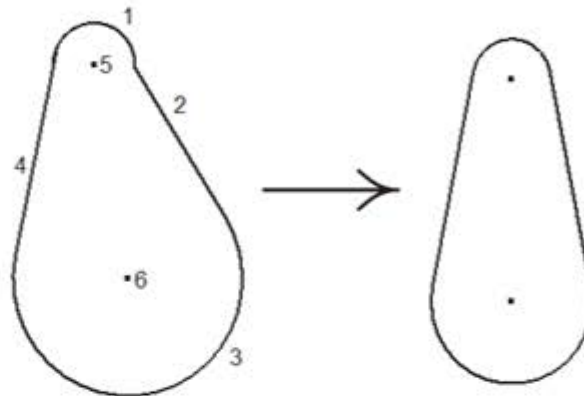


Figure 3–73

- 1 tangent with 2, 1 tangent with 4, 5 coincident with 6.
 - 1 tangent with 2, 1 tangent with 4, 5 concentric with 6.
 - 1 tangent with 2, 1 tangent with 4, 5 collinear with 6.
 - 1 tangent with 2, 1 tangent with 4, 5 vertical with 6.
11. Based on the constraint symbols shown in Figure 3–74, which constraint types are applied to the bottom horizontal entity? (Select all that apply.)



Figure 3–74

- Symmetric
- Parallel
- Perpendicular
- Tangent
- Collinear
- Coincident

12. Fillets and chamfers that are added in the sketch become separate Fillet and Chamfer features in the Model browser once the sketch has been completed. They can be selected and deleted independent of the sketch.
- a. True
 - b. False
13. What types of sketch entities can you use as the mirror line when mirroring geometry? (Select all that apply.)
- a. Construction line
 - b. Solid line
 - c. Projected part edge
 - d. Projected work plane

Chapter Review Questions

1. Which commands can be used to copy sketched entities? (Select all that apply.)
 - a. **Copy**
 - b. **Trim**
 - c. **Move**
 - d. **Extend**
2. Which of the following statements describes the **Move** command for sketch entities?
 - a. Continues an entity to meet the next entity in its path or to close an open sketch.
 - b. Enables you to relocate original 2D sketch geometry.
 - c. Removes the segment of the entity to the nearest intersection in each direction from the point selected.
 - d. Provides a specific tool for stretching sketched entities versus simply selecting and dragging them in the sketch.
3. You can rotate and make a copy of the entities at the same time.
 - a. True
 - b. False
4. Which of the following statements best describes the difference between the **Stretch** command and the select and drag technique for manipulating constrained geometry?
 - a. The **Stretch** command enables you to stretch constrained geometry, maintaining established constraints, whereas the select and drag technique cannot be used to manipulate constrained geometry.
 - b. The select and drag technique enables you to stretch constrained geometry, maintaining established constraints, whereas the **Stretch** command cannot be used to manipulate constrained geometry.

5. What is the purpose of defining the Base Point for the **Move**, **Copy**, **Scale**, and **Stretch** sketching tools?
 - a. The Base Point defines a point on the original geometry as the reference point for the operation.
 - b. The Base Point defines the location to which you are copying the selected entities.
6. You can copy a sketch from one part into another part.
 - a. True
 - b. False
7. Which of the following are required when creating a rectangular pattern in two directions, as shown in Figure 4–33? (Select all that apply.)

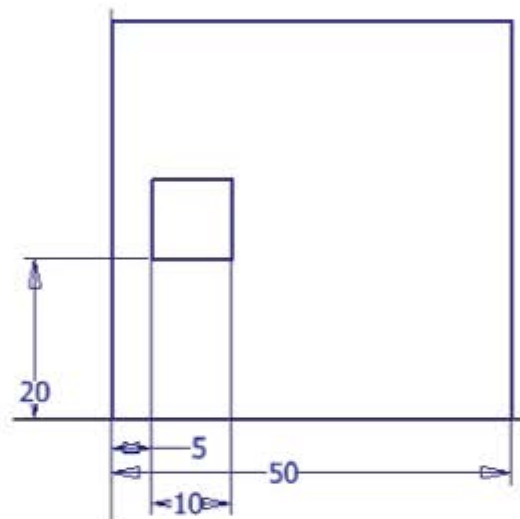


Figure 4–33

- a. Geometry to pattern (i.e., circular sketch)
 - b. First pattern direction
 - c. Second pattern direction
 - d. Angle between occurrences
 - e. Pattern axis
 - f. Count and Space in the first direction
 - g. Count and Space in the second direction
8. Which of the following tools enables you to customize the sizing of the grid display for a sketch?
 - a. Application Options
 - b. Document Settings

Chapter Review Questions

1. The **Offset** option is commonly used to create sketched entities offset from projected geometry. In the sketch shown in Figure 5–56, how many dimensions are required to fully constrain the sketch? Assume **Constrain Offset** and **Loop Select** are toggled on.

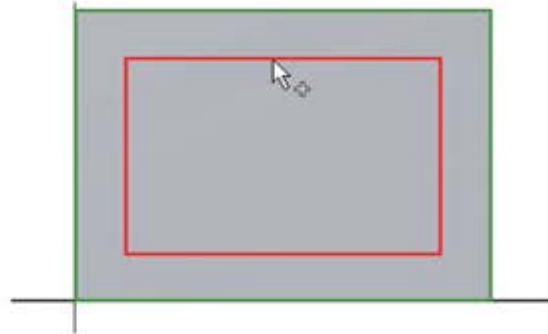


Figure 5–56

- a. 1
 - b. 2
 - c. 3
 - d. 4
 - e. More than 4
2. To create a new feature, you are required to reuse a sketch that was created for **Extrusion2**, as shown in Figure 5–57. Is this possible? If so, which option would you use?

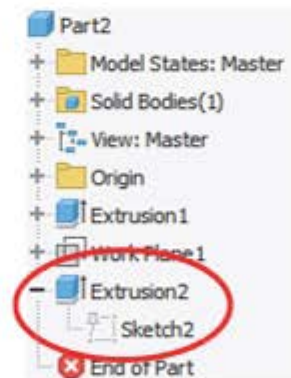


Figure 5–57

- a. You cannot reuse an existing sketch; you must recreate it using projected geometry.
- b. It is possible using the **Visibility** option.
- c. It is possible using the **Edit Sketch** option.
- d. It is possible using the **Share Sketch** option.

3. Which of the following is true regarding Projected Geometry in a sketch?
 - a. You can only project edges into a sketch that lie in the same plane as the sketching plane.
 - b. When selecting a Work Plane as a sketching plane, the edges that lie in the plane are automatically projected into the sketch.
 - c. When selecting an existing edge as a constraint or dimension reference, it is automatically projected into the sketch.
 - d. When you use a projected edge as a dimension reference, no relationship between the sketch and the projected edge's parent feature is established.
4. Which command is used to create an extrusion that adds material to existing solid geometry?
 - a.  (Join)
 - b.  (Intersect)
 - c.  (Cut)
 - d.  (Solid)
5. Which tool temporarily hides the portion of a part that is above the sketch plane to display the plane more clearly for sketching?
 - a. **Slice Graphics**
 - b. **Relax Mode**
 - c. **Look At**
 - d. **Hidden Edge Display**
6. In the Inventor Precise Input dialog box, which tool enables you to change the origin point so that it is used as the reference point from which values are measured?
 - a. 
 - b. 
 - c. 

7. The custom placement of the origin used with the Precise Input tool is only set as long as the entity that was being sketched at the time is active.
 - a. True
 - b. False
8. Which of the following statements are true regarding dynamic input and dimensioning when creating sketched entities? (Select all that apply.)
 - a. Dynamic input is available for all sketched entity types.
 - b. When dynamically entering the first dimension value (the locating dimension value) for an entity, the coordinate entry is always measured from (0,0,0).
 - c. By default, values entered in a dynamic input field automatically display as dimensions in the sketch.
 - d. Dynamic input cannot be disabled for sketched entities.

Chapter Review Questions

1. Which chamfer type requires you to select a reference surface?
 - a. **Distance**
 - b. **Distance and Angle**
 - c. **Two Distances**
2. In the fillet Tool palette, shown in Figure 6–84, which option enables you to fillet all of the edges of a feature?

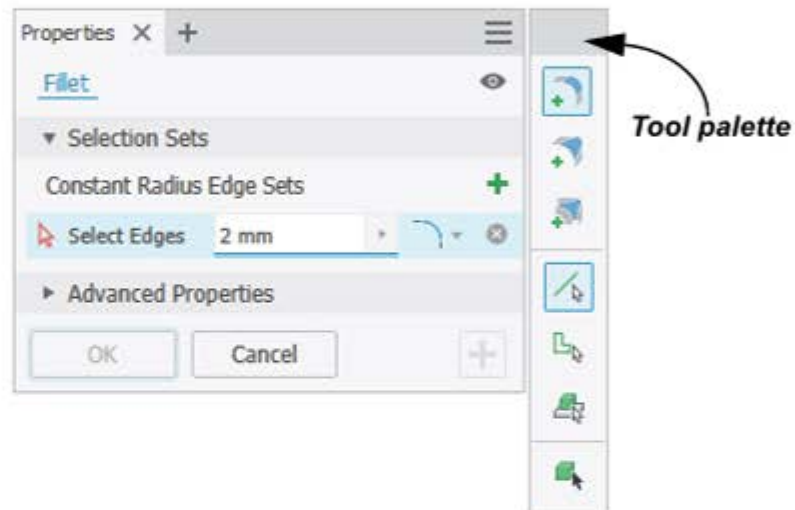


Figure 6–84

- a. 
 - b. 
 - c. 
3. Chamfer features always remove material.
 - a. True
 - b. False
 4. Fillet features can add or remove material.
 - a. True
 - b. False

5. Which keys on the keyboard can be used to clear the edges selected for a fillet or chamfer feature? (Select all that apply.)
 - a. <Tab>
 - b. <Ctrl>
 - c. <Shift>
 - d. <Alt>
6. Fillets on multiple edges can be controlled by one dimension.
 - a. True
 - b. False
7. How many separate Fillet features are listed in the Model browser if you click **OK** to create Fillet2, shown in the Properties panel in Figure 6–85?

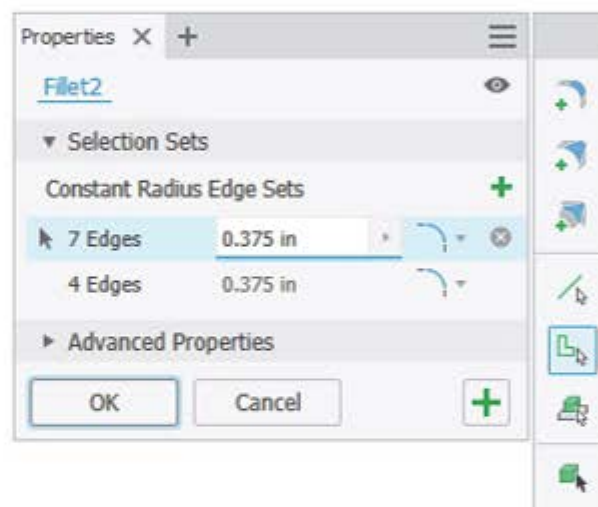


Figure 6–85

- a. 1
 - b. 2
 - c. 4
 - d. 11
8. You can add threads to both cylindrical faces and conical faces.
 - a. True
 - b. False
9. When defining the location reference for a concentric hole, which of the following can be selected?
 - a. Planar Face
 - b. Cylindrical Face

- | | |
|--|--|
| | <p>10. How can you modify an existing placed feature, such as a fillet or chamfer?</p> <ul style="list-style-type: none">a. Return to the sketch mode and change the sketch profile.b. Double-click on the feature in the model or in the Model browser.c. Select the same tool used to create the feature to edit it. <p>11. Which option do you select to change the reference edges for a chamfer?</p> <ul style="list-style-type: none">a. Show Dimensionsb. Edit Featurec. Edit Sketchd. Suppress Features |
|--|--|

Chapter Review Questions

1. Origin planes are the only work planes that can exist in a model.
 - a. True
 - b. False
2. Which of the following are possible uses for work planes similar to those shown in Figure 7–35? (Select all that apply.)

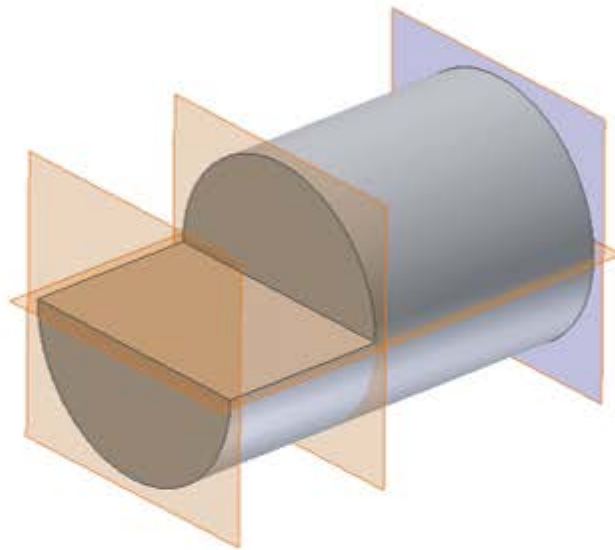


Figure 7–35

- a. To create another work plane.
 - b. To define a sketch plane.
 - c. To serve as the boundary for the depth of an extrusion.
 - d. To set up a drawing sheet.
3. Which of the following are valid options to create a work plane? (Select all that apply.)
 - a. **Through a Plane and Parallel to an Axis**
 - b. **Offset from Plane**
 - c. **Tangent to Surface and Parallel to Plane**
 - d. **Three Points**

4. Which option enables you to create the work plane shown in Figure 7–36?

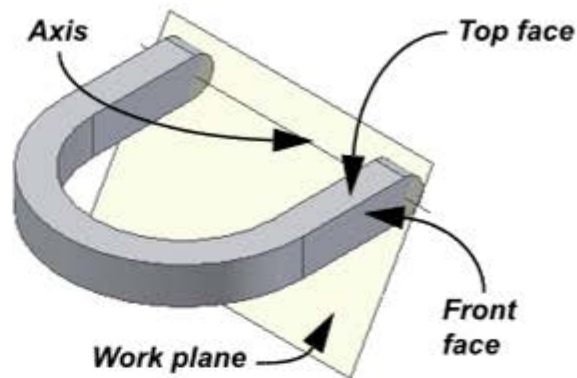


Figure 7–36

- a. **Tangent to Surface through Edge**
 - b. **Tangent to Surface through Point**
 - c. **Tangent to Surface and Parallel to Plane**
 - d. **Angle to Plane around Edge**
5. Which of the following are valid options for creating a work axis? (Select all that apply.)
- a. **Intersection of Two Planes**
 - b. **Through Two Points**
 - c. **Parallel to a Line**
 - d. **On Line or Edge**
6. A work axis can be used as a reference when creating a work plane.
- a. True
 - b. False
7. Which option creates a work point centered on a face and remains centered even after the size of the face changes?
- a. **On Vertex, Sketch Point, or Midpoint**
 - b. **Center Point of Loop of Edges**
 - c. **Center of Torus**
 - d. **Through Center of Circular or Elliptical Edge**
8. You can create work planes during work point creation to reference and constrain the work point.
- a. True
 - b. False

Chapter Review Questions

1. You can only add an equation to a model in the Parameters dialog box.
 - a. True
 - b. False
2. Which of the following statements regarding equations are true? (Select all that apply.)
 - a. Dimensions and parameters can be used in an equation to drive a value.
 - b. Equations can be manually entered in the Edit Dimension dialog box.
 - c. Equations can be created by selecting dimensions directly from the model.
3. Which combination of equations centers the hole if the length and width are equivalent, as shown in Figure 8–40? (Select all that apply.)

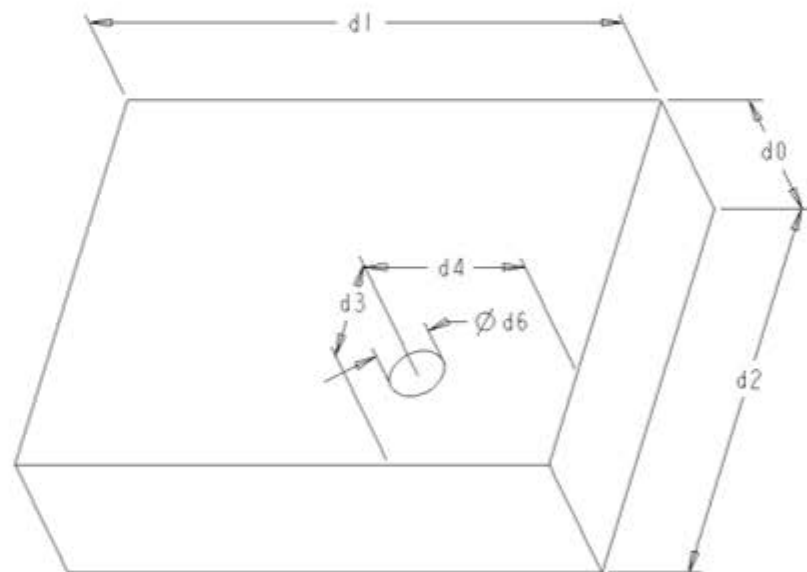


Figure 8–40

- a. $d4 = d1 / 2$
 $d3 = d2 / 2$
- b. $d1 = d4 / 2$
 $d2 = d3 / 2$
- c. $d1 = d2$
 $d4 = d1 / 2$
 $d3 = d2 / 2$

4. Which dimension display type is set to display the dimensions in the model shown in Figure 8–41?

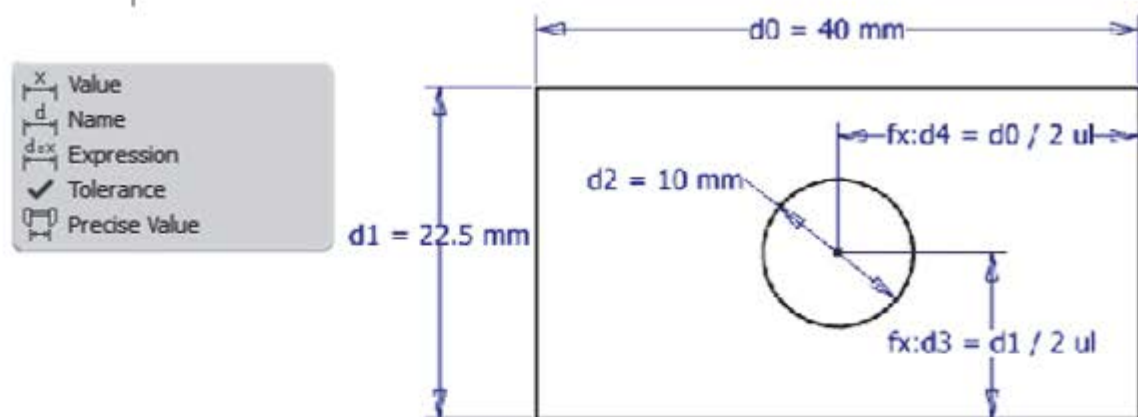


Figure 8–41

- Value
 - Name
 - Expression
 - Tolerance
 - Precise Value
5. Which description best describes how the equation shown in Figure 8–42 affects the model?

$$d0 = 3 * d6$$

$$d3 = d4$$

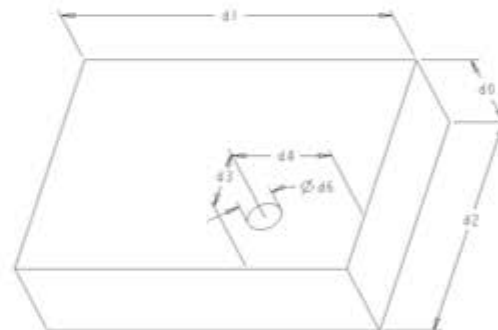


Figure 8–42

- The depth of the base extrusion is equal to the diameter of the hole and the hole is centered on the base extrusion.
- The depth of the base extrusion is equal to three times the diameter of the hole and the hole is centered on the base extrusion.
- The depth of the base extrusion is equal to three times the diameter of the hole, and the values of the horizontal and vertical hole dimensions are equivalent.
- The depth of the base extrusion is equal to three times the diameter of the hole, the values of the horizontal and vertical hole dimensions are equivalent, and d1 and d2 are equivalent.

6. Fill in the parameter type, **Model** or **User-Defined**, for the method in which it is created in a model.
- _____ parameters are created using the Parameters dialog box where you select the type, name, and enter values.
 - _____ parameters are automatically assigned each time you add a dimension or feature to a part.
7. Which of the following best describes what **fx** is identifying in the sketch shown in Figure 8–43?

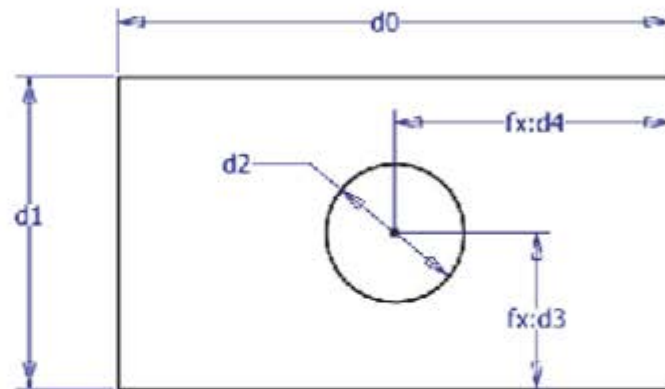


Figure 8–43

- d3 and d4 are reference dimensions.
- d3 and d4 are equal.
- d3 and d4 are generated based on a user-defined equation.
- d3 and d4 have a tolerance assigned.

Chapter Review Questions

1. Which of the following statements is true for creating a draft?
 - a. Multiple surfaces can be selected for drafting.
 - b. Multiple faces can be selected when defining the pull direction.
 - c. Only existing edges on the model can be selected to define the pull direction.
2. If the *Behavior* area of the split's Properties panel is displayed, is a face or a solid being split?
 - a. Face
 - b. Solid
3. You can shell the part without removing model faces.
 - a. True
 - b. False
4. Which of the following is true for shell features?
 - a. To create a shell, you must select the face(s) to remove.
 - b. A single face that is tangent to another face cannot be removed unless the tangent face is also removed.
 - c. A closed shell cannot have walls of varying thickness.
 - d. Wall thickness can be added to the inside or outside of a model.

5. When creating a shell, how can you leave one side of the part open, as shown in Figure 9–56?

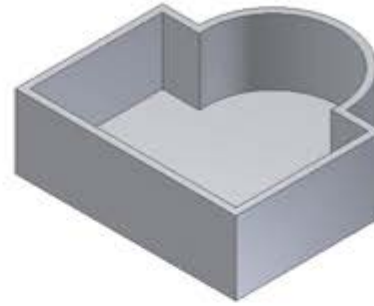


Figure 9–56

- a. Use **Remove Faces** in the Shell dialog box.
 - b. Set the *Thickness* for that face to **0**.
 - c. Delete the face after the shell has been created.
 - d. Set the *Thickness* for that face to the negative of the general thickness.
6. What can you do with the **Rib** command that you cannot do with the **Extrude** command?
- a. Create a surface object.
 - b. Cut through an existing solid.
 - c. Select multiple profiles for the feature.
 - d. Create a solid from an open profile.

Chapter Review Questions

1. In the example shown in Figure 10–28, **Hole1** is a **Through All** hole and **Extrusion2** is the solid cylindrical feature. What can you do to include **Extrusion2** in the **Hole1** feature (i.e., have the hole pass through the cylinder)?

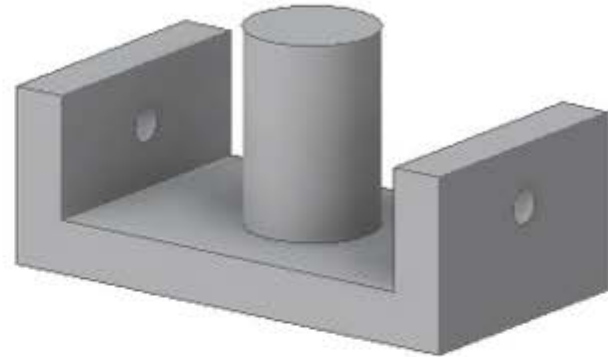
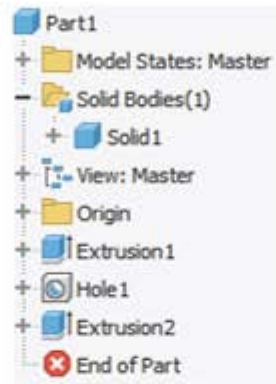


Figure 10–28

- a. Suppress **Hole1** and then recreate **Extrusion2**.
 - b. In the Model browser, drag the **End of Part** marker above **Hole1**.
 - c. Make **Extrusion2** Adaptive.
 - d. In the Model browser, drag **Extrusion2** above **Hole1**.
2. Which of the following statements are true for reordering features in a model? (Select all that apply.)
 - a. Reordering enables you to drag and drop features so that you can rearrange the feature creation order.
 - b. When reordering, features can be moved anywhere.
 - c. Base features can be moved anywhere in the model.
 - d. You can insert new features between existing features.
 3. Any features below the  **End of Part** marker in the Model browser are still displayed, but you cannot select them as references when creating new features.
 - a. True
 - b. False
 4. If you suppress a feature, all dependent features are also suppressed.
 - a. True
 - b. False

5. Which items in the Model browser shown in Figure 10–29 are *Suppressed* in the model? (Select all that apply.)

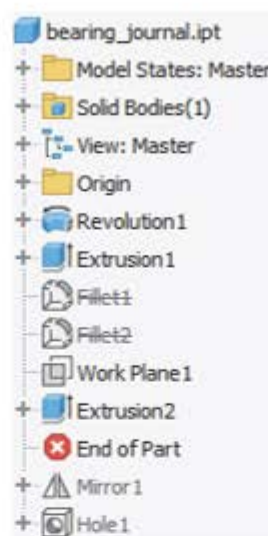


Figure 10–29

- a. Solid1
 - b. Sketch3
 - c. Fillet1
 - d. Fillet2
 - e. Work Plane1
 - f. Mirror1
 - g. Hole1
6. How can a section view be saved with the model?
- a. A section view is saved with the model and can be retrieved in the Navigation Bar.
 - b. A section view is saved with the model and can be retrieved in the Appearance panel in the *View* tab.
 - c. A section view is saved with the model in a design view and can be retrieved in the Model browser.
 - d. A section view cannot be saved with the model.
7. Once completed, a section view can be modified.
- a. True
 - b. False

8. Which of the following can be manipulated and stored in a design view? (Select all that apply.)
- a. Part color
 - b. Sketch and feature visibility
 - c. Display of Origin Planes
 - d. Section views
 - e. Model orientation
 - f. Model zoom level

Chapter Review Questions

1. The Design Doctor becomes active when a feature failure occurs during feature creation.
 - a. True
 - b. False
2. What happens when you add a dimension that over-constrains the sketch?
 - a. The dimension is added to the sketch.
 - b. The dimension is immediately added as a reference dimension.
 - c. The Sketch Doctor dialog box opens.
 - d. A dialog box opens indicating that the sketch is over-dimensioned.
3. Which of the following indicates a partially failed operation?
 - a. 
 - b. 
 - c. 
 - d. 
4. Which actions cause a sketch or feature to fail? (Select all that apply.)
 - a. Invalid sketch.
 - b. Too many sketch entities in the sketch.
 - c. Deleting a feature.
 - d. Modifying dimensions.
 - e. Deleting sketcher references.
 - f. Open loop sketch when creating a solid Extrude.
 - g. Cut not terminated by the termination plane.
 - h. Over-constrained sketch.

5. The failed feature can be the modified feature or a feature that references the modified feature.
 - a. True
 - b. False
6. What does the **red cross** button indicate if it displays in the Properties panel, as shown in Figure 11–19, while extruding a sketch?

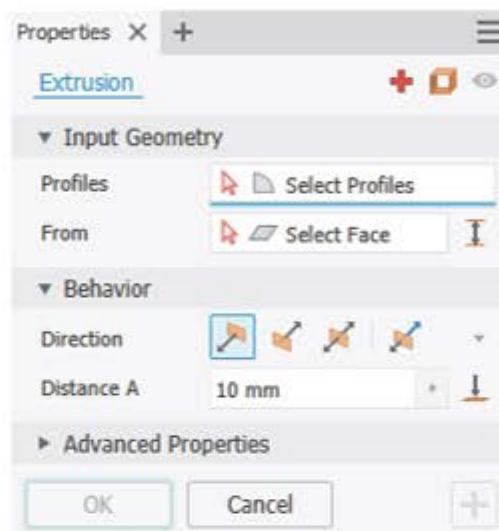


Figure 11–19

- a. The sketch might have a problem and the Sketch Doctor is available to help fix it.
 - b. There is applicable information available in the Help files about the **Extrude** command.
 - c. The sketch selected is not fully constrained.
 - d. The virtual memory on your computer might be too low for the operation.
7. When adding a dimension that is going to over-constrain the sketch, which options can you perform? (Select all that apply.)
 - a. Cancel the command.
 - b. Accept the warning and place a driven dimension.
 - c. Place the same dimension twice.
 - d. Replace the conflicting dimension with the new one.

Chapter Review Questions

1. Which command would you use to create the groove in the model shown in Figure 12–22?

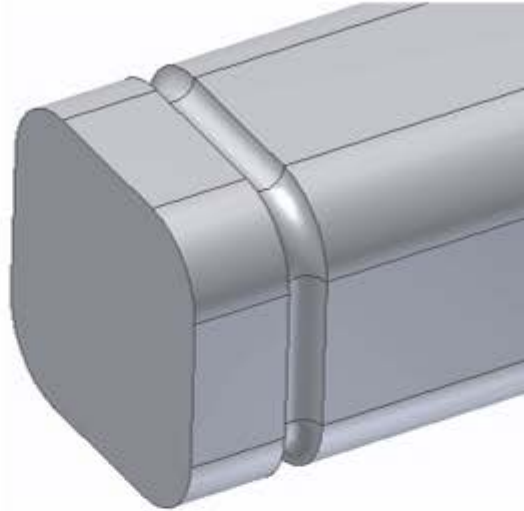


Figure 12–22

- a. **Coil**
 - b. **Revolve**
 - c. **Extrude**
 - d. **Sweep**
2. Which elements must be defined before starting the creation of a Sweep feature? (Select all that apply.)
 - a. Profile
 - b. Path
 - c. Section
 - d. Start Point
 3. A Sweep feature can be used to either add or remove material.
 - a. True
 - b. False

4. Which of the following statements for Sweep features is true?
 - a. A sweep creates a single feature whose geometry is blended between multiple profiles.
 - b. A sweep creates a single feature whose geometry is swept along a defined path.
 - c. A sweep can only be added to the model after the base extrusion has been created.
 - d. A sweep path must be sketched geometry.
5. You can taper a sweep with a closed path, but you cannot taper a sweep with an open path.
 - a. True
 - b. False
6. You can twist the path that defines a sweep by entering a Twist value.
 - a. True
 - b. False
7. How many unconsumed sketches are required to create the sweep feature shown in Figure 12–23?

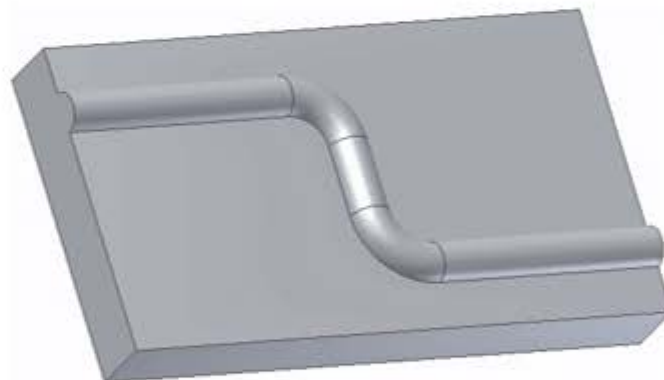


Figure 12–23

- a. 1
- b. 2
- c. 1 for 2D sweep, 2 for 3D sweep.
- d. A sweep is not based on a sketch.

Chapter Review Questions

1. A loft can be used in a model to either add or remove material.
 - a. True
 - b. False
2. Which of the following best describes the type of geometry that is created using a Loft feature?
 - a. A loft creates multiple solid features between each of the profiles.
 - b. A loft creates a feature where the geometry is blended between multiple profiles.
 - c. A loft creates a feature whose geometry is swept along a defined path.
 - d. A loft creates a feature that is rotated around a selected centerline.
3. Which of the following statements for Loft features are true? (Select all that apply.)
 - a. A loft can only be added to the model after the base extrusion has been created.
 - b. The profiles for lofts can only be sketched.
 - c. Once you select the profiles for the Loft feature, you cannot reorder them.
 - d. To use an existing planar face as a section of a loft, you can select the face directly without creating a sketch.
 - e. Map points are used to determine the twist of the loft.
4. Which type of loft can be used to create a loft that passes through the three circular sections and is guided along the curve, as shown in Figure 13–44?

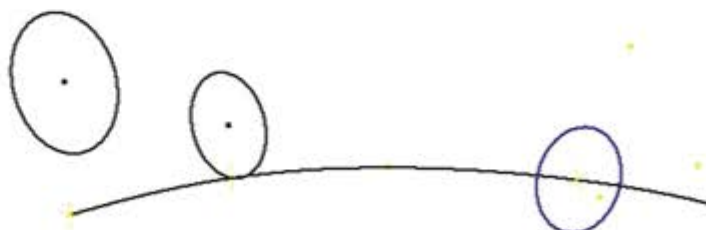


Figure 13–44

- a. Rail
- b. Center Line

5. When referencing existing geometry as the start or end section for a loft, which of the following conditions provides access to the *Weight* settings to control its shape, as shown in Figure 13–45? (Select all that apply.)

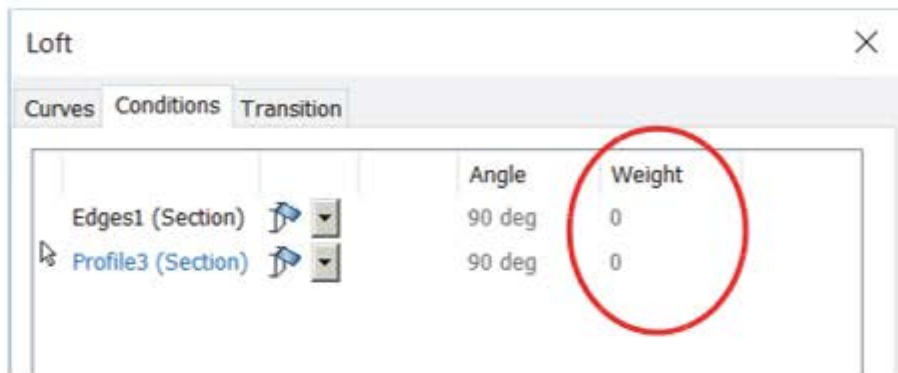


Figure 13–45

- a. Free
- b. Tangent
- c. Smooth (G2)

Chapter Review Questions

1. The direction references selected to create a rectangular feature pattern must be linear.
 - a. True
 - b. False
2. The model shown in Figure 14–45 has a pattern of extruded cuts created in one direction. What is the number of pattern occurrences that are defined in the dialog box?

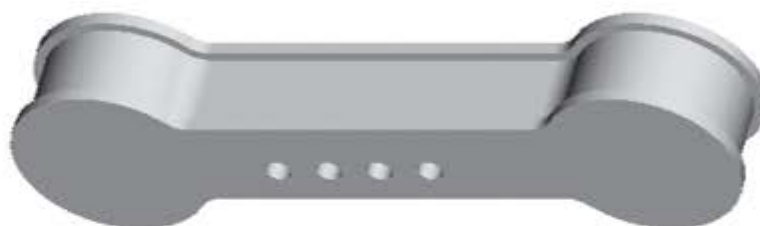


Figure 14–45

- a. 1
 - b. 2
 - c. 3
 - d. 4
3. Which entities can be used as the rotation axis for a circular pattern? (Select all that apply.)
 - a. Work plane
 - b. Edge
 - c. Work axis
 - d. Cylindrical face
 4. What should you do if the preview of your rectangular pattern is in the wrong direction along a selected path? (Select all that apply.)
 - a. Select a new direction reference that is in the correct direction.
 - b. Click the **Flip** direction icon in the Rectangular Pattern dialog box to change the direction.
 - c. Enter a negative value as the *Spacing* value.
 - d. Expand the additional options at the bottom of the dialog box and change the Compute method.

5. The sketch that defines the pattern points in a Sketch Driven Pattern must be created after the **Sketch Driven Pattern** command is initiated.
 - a. True
 - b. False
6. Which of the following statements is true for removing specific occurrences from within a pattern?
 - a. You can suppress the first occurrence of the pattern (original).
 - b. You can suppress all but the first occurrence of the pattern.
 - c. You can select individual occurrences in the pattern for deletion.
 - d. You can use the **Delete** command to delete the first occurrence in the model.
7. Which of the following are valid characteristics of patterned features when using an Identical Compute method, as shown in Figure 14–46? (Select all that apply.)



Figure 14–46

- a. The created features cannot lie on different surfaces.
 - b. The created features can break edges of the part.
 - c. The created features cannot intersect with each other.
 - d. Feature depth of all occurrences is the same value as the pattern leader (regardless of the depth option set for the feature).
8. Which work feature can be used as a reference for mirroring features?
 - a. Work plane
 - b. Work point
 - c. Work axis

9. You can mirror an entire solid or individual features in a model.
- a. True
 - b. False

Chapter Review Questions

1. A pick and place feature can be the parent of another pick and place feature.
 - a. True
 - b. False
2. Which of the following actions create a feature relationship when sketching a feature? (Select all that apply.)
 - a. Selecting the sketching plane.
 - b. Selecting sketching references.
 - c. Creating an offset entity.
 - d. Using an existing edge to create a new entity.
 - e. Dimensioning the length of a line.
 - f. Aligning a new entity with an existing entity.
 - g. Extruding to a Blind depth.
 - h. Extruding through all surfaces.
 - i. Extruding to intersect with a selected face or plane.
3. Which of the following features create a parent/child relationship when created in a model? (Select all that apply.)
 - a. Origin Planes
 - b. Work Planes
 - c. Extrude
 - d. Hole
 - e. Chamfer
 - f. Fillet
 - g. Shell
4. Which of the following processes of creating a sketched cut feature result in the creation of feature relationships? (Select all that apply.)
 - a. Selecting sketch planes.
 - b. Selecting sketching references.
 - c. Cutting through all geometry.

5. If a parent feature is deleted, which of the following options can be used to change the references to the child feature? (Select all that apply.)
- a. **Edit Sketch**
 - b. **Share Sketch**
 - c. **Redefine**
 - d. **Edit Feature**
6. The Delete Features dialog box opens if you select and delete the hole shown in Figure 15–21. Which of the following best describes why this dialog box becomes available?

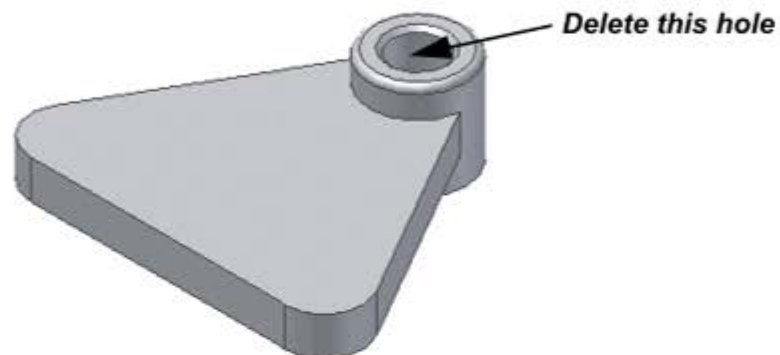


Figure 15–21

- a. Because the Delete Features dialog box opens when deleting all of the pick and place feature types.
 - b. Because a chamfer was placed on the edge generated by the creation of the hole.
 - c. Because there are other pick and place features in the model and you need to select the other ones to delete at the same time.
 - d. Because the hole was placed as a Coaxial hole and is a child to the extruded cylinder.
7. The  (Make Selected) icon in the Relationships dialog box only enables you to change the active feature to a listed parent feature. You cannot use it to activate a child.
- a. True
 - b. False

Chapter Review Questions

1. You cannot assemble components so that two parts occupy the same space at the same time.
 - a. True
 - b. False
2. In the Assembly browser, shown in Figure 16–54, what does the Pushpin icon next to the **Base_vise** part indicate?

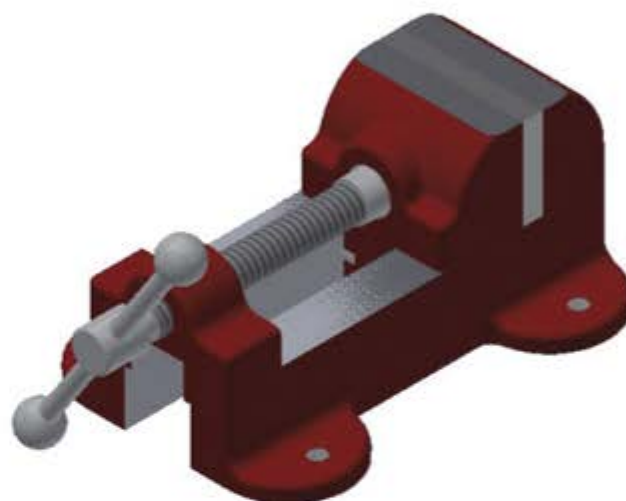
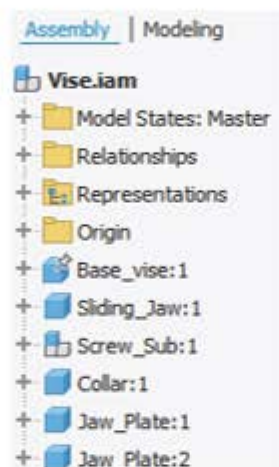


Figure 16–54

- a. The component requires additional constraints.
 - b. The component is adaptive (size adjusts to fit constraints).
 - c. The component was imported in SAT format (not parametric).
 - d. The component is grounded (does not move).
3. How many rotational degrees of freedom (R) and translational degrees of freedom (T) does an unconstrained part or subassembly have?
 - a. 1 R and 3 T
 - b. 3 R and 3 T
 - c. 2 R and 2 T
 - d. 3 R and 1 T

4. In the Place Constraint dialog box, four areas are circled, as shown in Figure 16–55. Which area enables you to assign a constraint between two parallel surfaces as **Flush**?

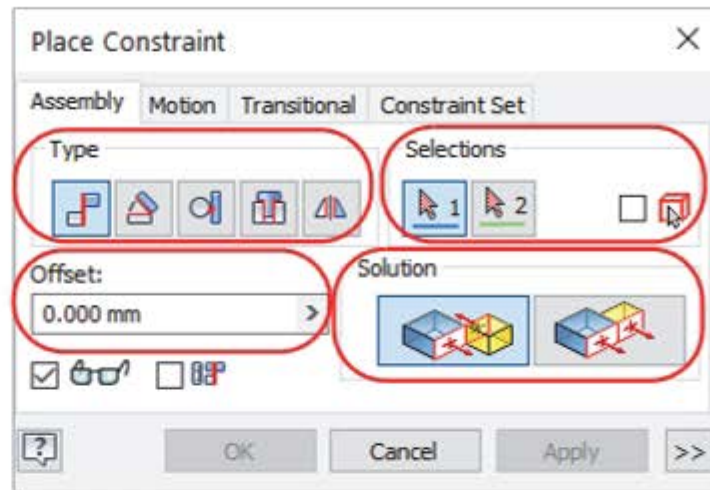


Figure 16–55

- Type
 - Solution
 - Offset
 - Selections
5. Which of the following constraint types is highlighted in the Place Constraint dialog box, as shown in Figure 16–56?

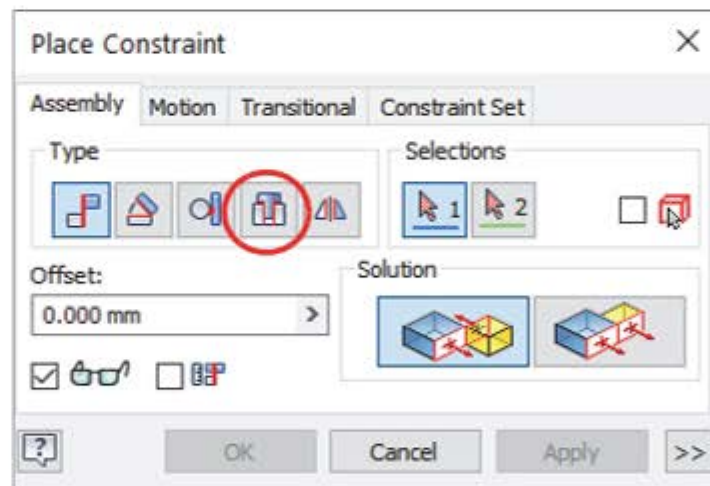


Figure 16–56

- Pin
- Flush
- Insert
- Mate

6. How would you determine how many degrees of freedom exist on a partially constrained part?
 - a. All degrees of freedom are automatically displayed in the model when the Place Constraint dialog box is open.
 - b. Right-click in the graphics window and select **Degree of Freedom** in the marking menu.
 - c. In the *Assemble* tab>Productivity panel, expand the commands and click  (Degree of Freedom Analysis).
 - d. Degrees of Freedom can only be reviewed in a sketch, not in an assembly.
7. Which of the following constraint combinations can be used to create the assembly shown on the right in Figure 16–57?

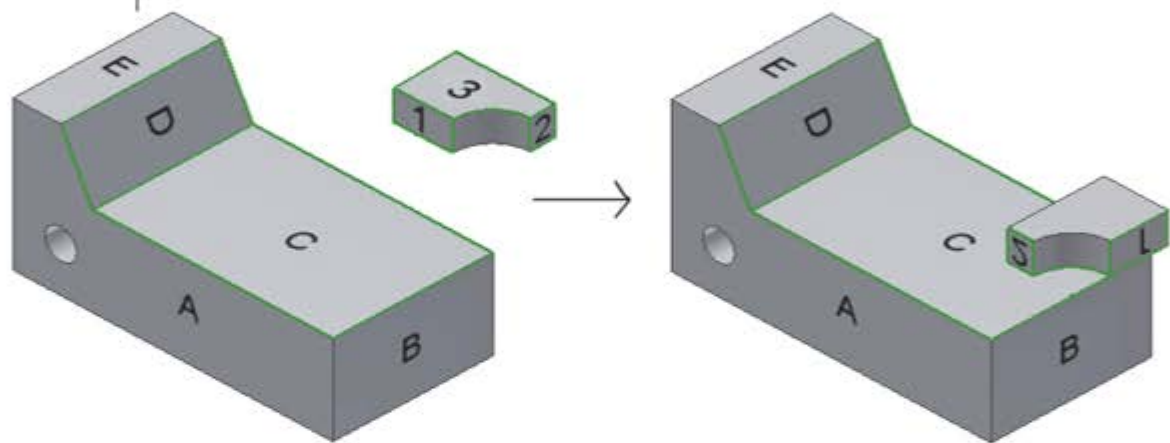


Figure 16–57

- a. Flush 1 and B, Flush Offset A and 2, Flush C and 3.
- b. Mate 1 and B, Mate Offset A and 2, Flush C and 3.
- c. Flush 1 and B, Flush Offset A and 2, Mate C and 3.
- d. Flush 1 and B, Flush Offset A and 2, Insert C and 3.
- e. Mate 1 and B, Mate Offset A and 2, Insert C and 3.
- f. Mate 1 and B, Mate Offset A and 2, Mate C and 3.

8. The pin is constrained in the holes on the part, as shown in Figure 16–58. How would you precisely control the distance between the pin heads and the sides of the part?

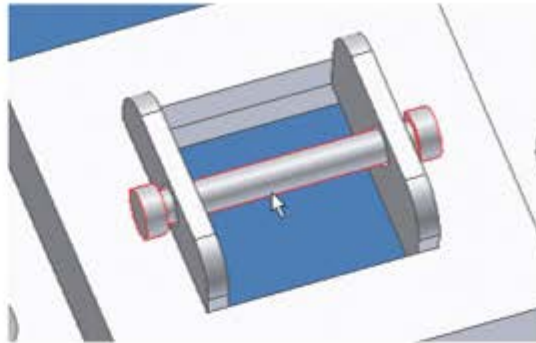


Figure 16–58

- a. Set the offset value for the constraint.
 - b. Use **Move Component**.
 - c. Use **Tweak Component**.
 - d. Apply a model dimension in the assembly.
9. Which of the following statements are true of subassemblies? (Select all that apply.)
- a. They are always grounded in the assembly.
 - b. They can contain other subassemblies.
 - c. They are always stored as .IAM files.
 - d. They act like a single component in the assembly.
10. What happens when you use the **Place** command to place an assembly file?
- a. It becomes a subassembly in the assembly.
 - b. The parts of the assembly are placed as individual components at the top level of the assembly.
 - c. It becomes a grounded component in the assembly no matter how many components have been assembled before it.
 - d. The parts of the assembly are placed but are not enabled.

11. Which constraint type creates a Flush constraint, as shown in Figure 16–59?

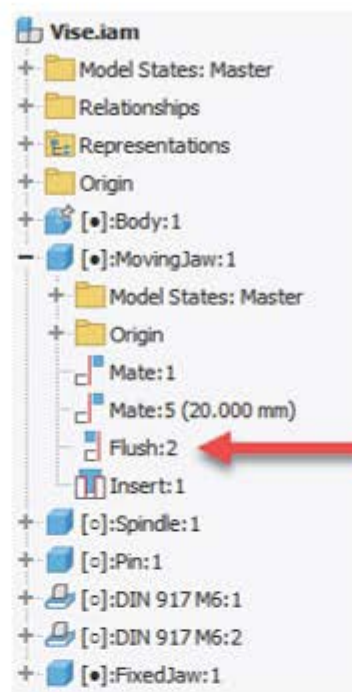


Figure 16–59

- a. Insert
 - b. Tangent
 - c. Mate
 - d. Angle
12. What type of entities can be constrained with the Insert constraint?
- a. Cylindrical entities
 - b. Linear entities
 - c. Flat entities
 - d. Points
13. When you set the Model browser to **Assembly View**, which of the following are listed in the Model browser display? (Select all that apply.)
- a. Origin features
 - b. Assembly constraints
 - c. Assembly features
 - d. Part features

14. For the software to automatically set the size of a Content Center model, the **AutoDrop** option must be toggled off.
- a. True
 - b. False

Chapter Review Questions

1. Which of the following statements are true about moving components in an assembly? (Select all that apply.)
 - a. Components that are partially constrained can be moved by selecting and dragging them.
 - b. Components that are fully constrained can be moved using the **Free Move** command.
 - c. Components that are partially constrained can be moved using the **Free Move** command.
 - d. Components that are fully constrained can be moved by selecting and dragging them.
2. If you move constrained components apart (as shown in Figure 18–24), how can you display them in their constrained positions again?

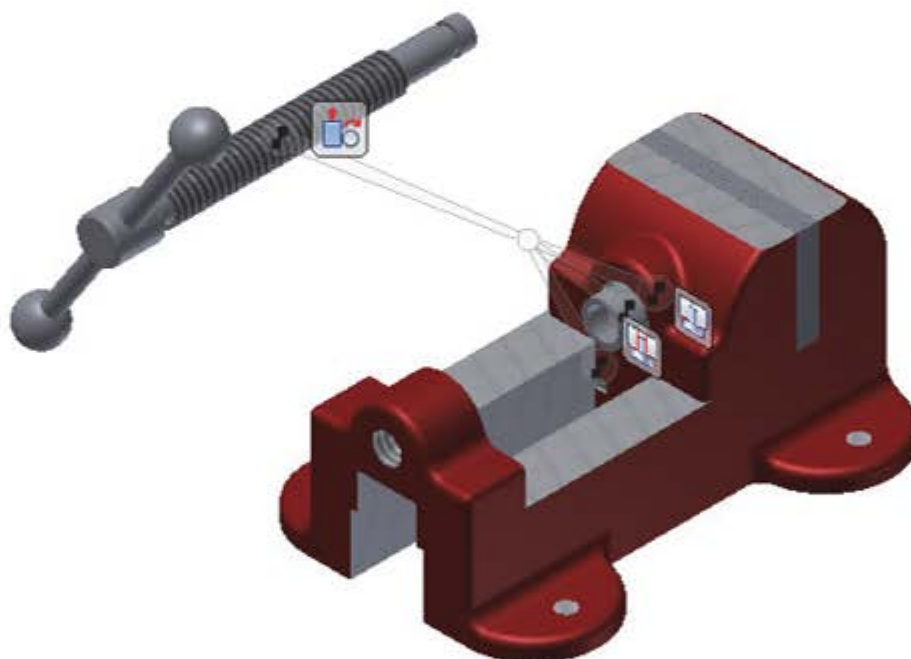


Figure 18–24

- a. Use **Update**.
- b. Use **Degrees of Freedom** in the *View* tab.
- c. Use **Zoom All**.
- d. Reapply the constraints.

3. Which of the following statements about  (Free Rotate) is true?
- You can rotate multiple components at the same time.
 - Only unconstrained components can be rotated.
 - Grounded components cannot be rotated.
 - When you rotate a component that is constrained, the constraints change based on a new component position.
4. Which of the following statements is true regarding suppressing constraints, as shown in Figure 18–25?

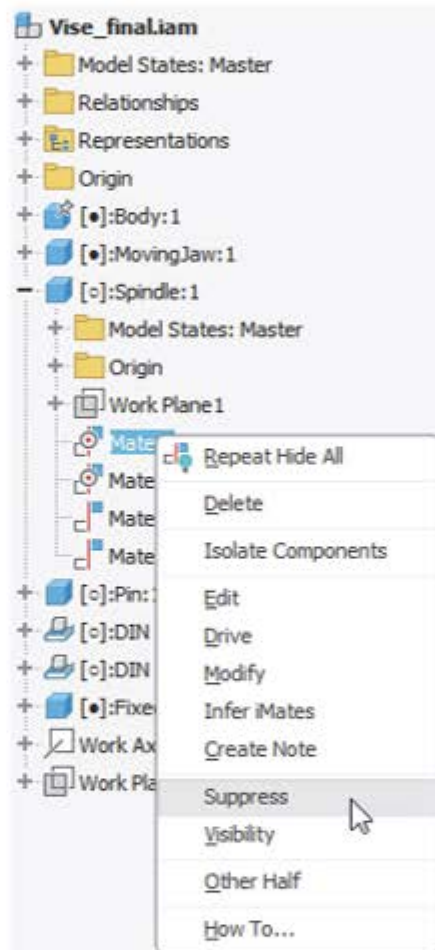


Figure 18–25

- Suppressing a constraint removes both components referenced in the constraint from the display.
- Suppressing a constraint locks the constraint references such that they cannot be edited.
- Suppressing a constraint removes its associated component from the display.
- Suppressing a constraint enables you to ignore the constraint restrictions and drag the component as required in the assembly.

5. What is the purpose of a section view?
 - a. To simplify the display of an assembly.
 - b. To update an assembly more quickly.
 - c. To create an exploded view of an assembly.
 - d. To work inside an assembly while keeping the required components visible.
6. What is the purpose of a design view? (Select all that apply.)
 - a. To simplify the display of an assembly.
 - b. To update an assembly more quickly.
 - c. To return to a specific view of an assembly.
 - d. To rotate assembly components.
7. Which of the following filter options is used in the Edit View mini-toolbar to remove components from an existing design view?
 - a. **View All**
 - b. **View Included**
 - c. **View Excluded**
8. When a section view is active (as shown in the Three Quarter Section View in Figure 18–26), you cannot select components or reference entities.

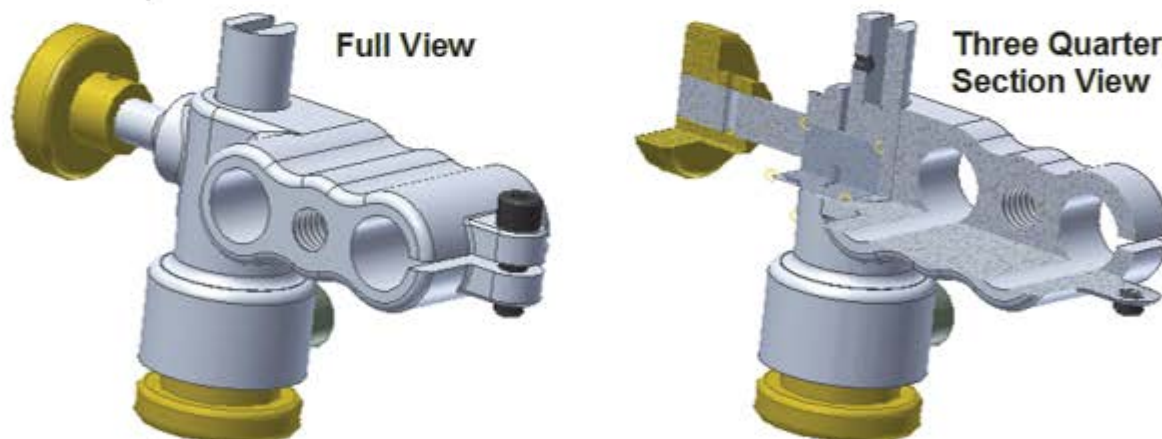


Figure 18–26

- a. True
- b. False

9. What is the difference between toggling off the visibility of a component and isolating that same component?
 - a. When the visibility of a component is off, it is not displayed. However, when isolated, the visibility of all of the other components is toggled off and only the selected component is visible.
 - b. No difference. Either option can be used to remove the component from the display.
 - c. When the visibility of a component is off, it is not displayed; however, when isolated, the component is activated for editing.
10. Which of the following statements are true for the Model browser and assembly shown in Figure 18–27? (Select all that apply.)

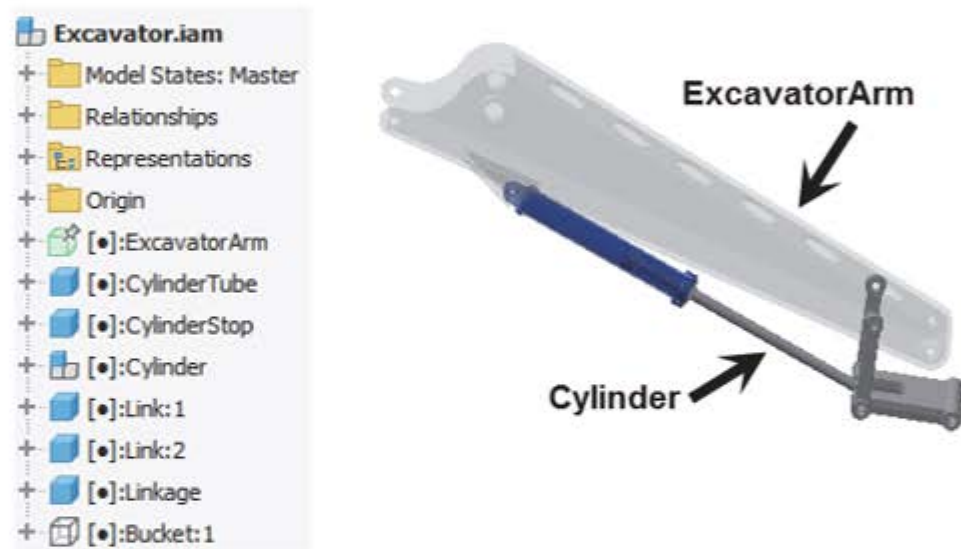


Figure 18–27

- a. **Bucket's** visibility has been cleared.
 - b. **Excavator Arm** and **Cylinder** have been enabled.
 - c. **Excavator Arm** is a grounded component.
 - d. **Excavator Arm** has been isolated.
11. Which of the following statements is true of a part that is disabled (i.e., the **Enabled** option is cleared) in an assembly?
 - a. It is completely invisible.
 - b. It is completely visible but not editable.
 - c. It is displayed as transparent and is not selectable.
 - d. It is not subjected to constraints.

Chapter Review Questions

1. Which of the following measurements can be generated using the **Measure** option? (Select all that apply.)
 - a. Edge Length
 - b. Diameter
 - c. Angle
 - d. Distance between points
 - e. Distance between edges
 - f. Area of a sketch
2. Which of the following best describes how to use the **Measure** command to enter a required value when creating a feature?
 - a. With the Properties panel, dialog box, or mini-toolbar open, in the *Inspect* tab>Measure panel, select **Measure**. Select the geometrical references to perform the required measure. This value is automatically used in the value field.
 - b. In the Properties panel, dialog box, or mini-toolbar, click the arrow next to the value field you want to change and select **Measure** in the drop-down list. Select the geometrical references to perform the required measure. This value is automatically used in the value field.
 - c. In the *Inspect* tab>Measure panel, select **Measure**. Select the geometrical references to perform the required measure. Create the feature and in the expanded menu for the required dimension value, select the value that was just measured.
3. The iProperties dialog box provides you with information about which physical properties of a part? (Select all that apply.)
 - a. Mass
 - b. Center of Gravity
 - c. Area
 - d. Volume

4. In an assembly file, the material cannot be changed using the iProperties dialog box.
 - a. True
 - b. False
5. Where do you find information such as the mass and volume of a part, as shown in Figure 19–27?

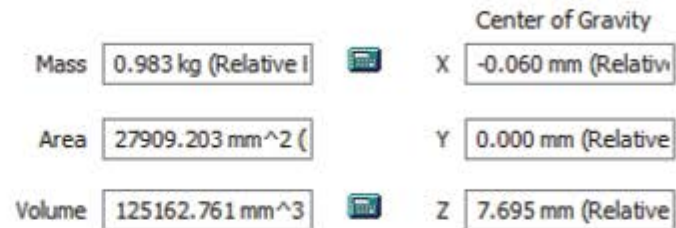


Figure 19–27

- a. Analysis Tools (*Inspect* tab>Measure panel)
 - b. iProperties dialog box (**File** menu>**iProperties**)
 - c. Parameters dialog box (*Manage* tab>**Parameters**)
 - d. Document Settings dialog box (*Tools* tab>**Document Settings**)
6. Which of the following best describes how to change the color of a specific feature in the model?
 - a. Select the feature and select a material that uses the appropriate color in the Materials drop-down list.
 - b. Select the feature and select a material that uses the appropriate color in the *Physical* tab in the iProperties dialog box.
 - c. Select the feature and select a color in the Appearance Override drop-down list.
 - d. Select the feature and select a color in the Properties dialog box.

Chapter Review Questions

1. Which of the following file formats is used to create an exploded assembly model in a Drawing view?
 - a. .IAM
 - b. .IPT
 - c. .IPN
 - d. .DWG
2. What is the purpose of a presentation file? (Select all that apply.)
 - a. To simplify the display of an assembly.
 - b. To create an Exploded View of an assembly.
 - c. To update an assembly more quickly.
 - d. To create an animation of an assembly.
 - e. To help document and visualize the assembly.
3. Which of the following statements are true regarding a presentation file? (Select all that apply.)
 - a. Once the Presentation template is selected for use, you are immediately prompted to open the model that will be used in **Scene1** of the presentation.
 - b. A Snapshot view that is oriented in the model's Home view is automatically added to a new presentation file.
 - c. Multiple storyboards can be created in a presentation file to document an assembly.
 - d. Storyboard animations can be used in a drawing view.
4. It is not possible to edit a Snapshot view that was created dependent on a specific time in an animation.
 - a. True
 - b. False
5. Which command enables you to save a specific view orientation at a set time in an animation?
 - a. **New Storyboard**
 - b. **New Snapshot View**
 - c. **Tweak Components**
 - d. **Capture Camera**

6. What is the purpose of adding a trail?
 - a. To define a path for an animation.
 - b. To move the position of a component.
 - c. To change the color of a component.
 - d. To help define the relationships between the components in terms of how they are assembled.
7. To create an animated assembly of a model's assembly process, you must create an animation that uses the _____ command.
 - a. **New Storyboard**
 - b. **Tweak Components**
 - c. **Opacity**
 - d. **Capture Camera**
8. Which of the following are valid methods to change the duration of an action in a storyboard animation? (Select all that apply.)
 - a. Enter a *Duration* value in the mini-toolbar during Tweak creation.
 - b. Use the **Edit Tweak** command and enter a new *Duration* value.
 - c. Use the **Edit Time** command and enter a new *Duration* value.
 - d. Drag the action's duration directly in the timeline.
9. **Move** and **Rotate** tweaks can be assigned to the same component at one time.
 - a. True
 - b. False
10. Which type of view setting can only be instantaneous when assigned to an animation?
 - a. Opacity
 - b. Visibility

Chapter Review Questions

1. When replacing a part in the assembly, you can replace all instances of the part at the same level of the assembly.
 - a. True
 - b. False
2. What is the effect of promoting a component (as shown in Figure 21–39) in an assembly?

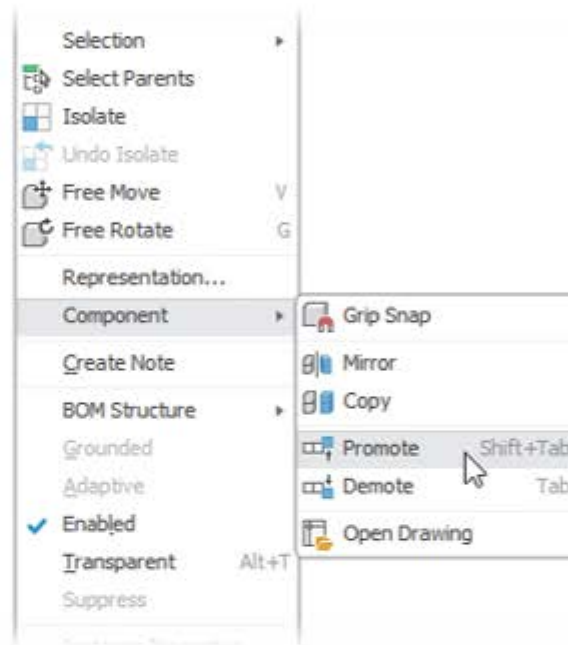


Figure 21–39

- a. It makes that component active for editing.
 - b. It moves it from a subassembly into the next higher level assembly.
 - c. It makes it a separate part file.
 - d. It makes it the grounded component.
3. You can use the **Demote** option to move components into an existing subassembly.
 - a. True
 - b. False

4. Which of the following best describes the purpose of a Drive constraint?
- a. A Drive constraint is a dimension type that is created in a sketch of an assembly feature so that the value can be easily changed to update the assembly.
 - b. A Drive constraint is the terminology used for the full set of constraints to fully locate a component in an assembly.
 - c. A Drive constraint enables you to assign motion between the constrained components by setting the offset or angle value in a constraint so that it changes incrementally.
 - d. A Drive constraint is a type of constraint that is assigned in the Place Constraint dialog box.
5. Which of the following components have been specified as members in a contact set, as shown in Figure 21–40? (Select all that apply.)

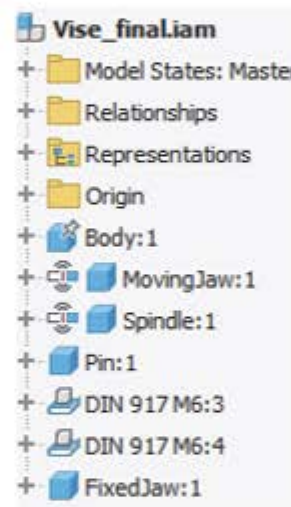


Figure 21–40

- a. Body
 - b. Moving Jaw
 - c. Spindle
 - d. Pin
 - e. DIN 917 M6:3
 - f. DIN 917 M6:4
 - g. Fixed Jaw
6. You cannot assemble components so that two parts occupy the same space at the same time.
- a. True
 - b. False

7. Which of the following determines where two selected assembly components overlap?
- a. In the *Assemble* tab>Productivity panel, expand the commands and click .
 - b. In the *Inspect* tab>Interference panel, click .
 - c. *Inspect* tab>Measure panel, click .
 - d. An interference analysis can only be done on the whole model, not individual components.
8. What information can you find in the Interference Detected dialog box? (Select all that apply.)
- a. Number of interferences
 - b. Total Volume of interference
 - c. Part names
 - d. Subassembly names
9. Which of the following statements regarding the **Design Doctor** tool is true?
- a. It toggles the visibility of the component.
 - b. It displays options to resolve a constraint issue.
 - c. It enables a component so that it can be selected.
 - d. It moves a component out of the subassembly.

Chapter Review Questions

1. Which of the following view types must be the first view in a drawing?
 - a. Projection
 - b. Base
 - c. Auxiliary
 - d. Detailed
2. If you have already placed a front view of a part in a drawing, which tool most easily creates a side view aligned with the front view, as shown in Figure 25–59?

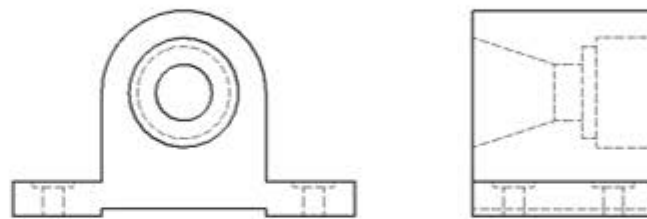


Figure 25–59

- a. Projected View
 - b. Base View
 - c. Draft View
 - d. Aligned View
3. Which command can create the foreshortening effect in the views shown in Figure 25–60?

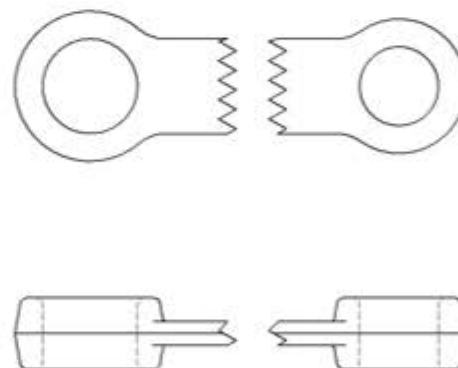


Figure 25–60

- a. Break View
- b. Crop View
- c. Auxiliary View
- d. Break Out View

4. Which view types enable you to create an independent 3D view? (Select all that apply.)
 - a. Projection
 - b. Base
 - c. Auxiliary
 - d. Detailed
5. You can break the alignment dependency between a parent and a child view.
 - a. True
 - b. False
6. If you change a part or assembly file, drawing views that show it update automatically by default.
 - a. True
 - b. False
7. When inserting a Base view into a drawing file, what is required to display a view name, similar to that shown in Figure 25–61?

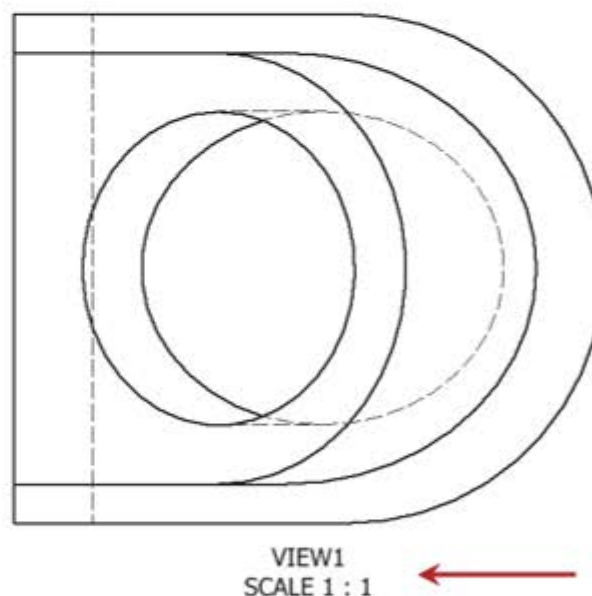


Figure 25–61

- a. Enable/Disable feature preview.
- b. Toggle the label **Visibility**.
- c. Ensure that you have edited the view label.
- d. Ensure that the **Scales** are on.

8. Which type of file is required to be created when you want to show an exploded view of an assembly in a drawing file, as shown in Figure 25–62?

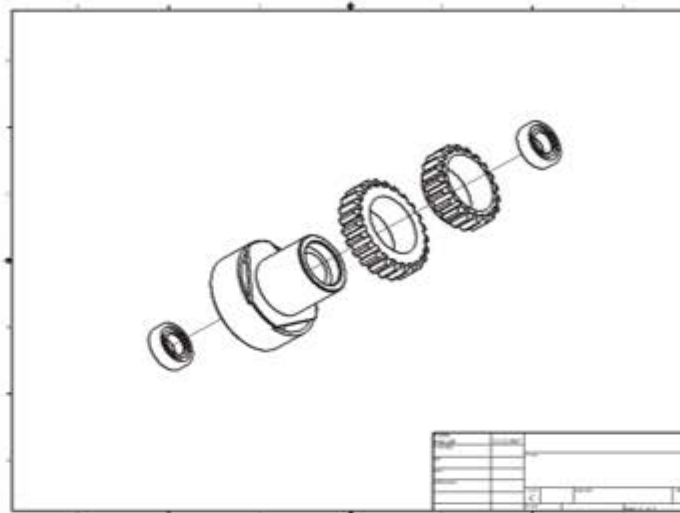


Figure 25–62

- a. Project
 - b. Part
 - c. Presentation
 - d. Assembly
9. When you delete a view, as shown in Figure 25–63, what happens to the dependent views?

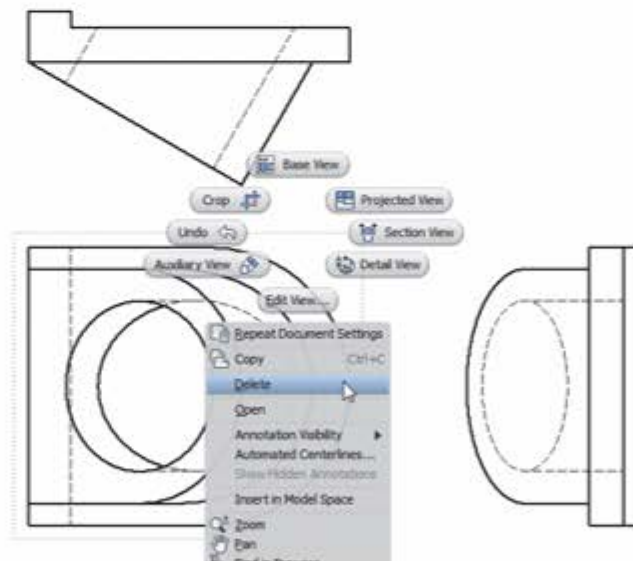


Figure 25–63

- a. They are automatically deleted.
- b. The dependent views are suppressed.
- c. You cannot delete a view that has dependent views.
- d. You are prompted to delete its dependent views.

10. A shaded base view exists in a drawing. This base view is automatically displayed as shaded and it cannot be modified to remove the shaded display.
 - a. True
 - b. False
11. How can you move a base view and its dependent views?
 - a. Move one of the dependent views and the base view moves with it.
 - b. Move the base view and dependent views individually.
 - c. Move the base view then realign the dependent views.
 - d. Move the base view and the dependent views move with it.
12. Which of the following can be changed when you edit a view? (Select all that apply.)
 - a. View Scale
 - b. Line type
 - c. View Label
 - d. Hidden line visibility
13. You can change the scale of the projected views independent of the parent view scale.
 - a. True
 - b. False

Chapter Review Questions

1. Which describes the difference between Model dimensions and Drawing dimensions?
 - a. Model dimensions can be edited to change the part, while Drawing dimensions are descriptive only and cannot be edited directly.
 - b. There is no difference, except whether you are viewing the dimensions in the model or in the drawing.
 - c. Model dimensions only display in part files, while Drawing dimensions display in drawing views.
 - d. Model dimensions cannot be deleted, but Drawing dimensions can be deleted.
2. How would you display model dimensions in an existing view where they are not showing?
 - a. Use General Dimension.
 - b. Right-click on the view and select **Retrieve Model Annotations**.
 - c. Update the view.
3. You can display the same model dimension in multiple views at the same time.
 - a. True
 - b. False
4. Which of the following best describes the **Ordinate Set** command, as shown in Figure 26–67?

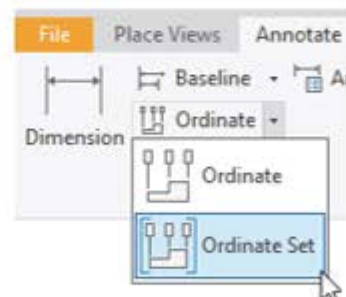


Figure 26–67

- a. It sets the 0,0 point to be used with Ordinate Dimensions.
- b. It creates a group of dimensions that act as one unit.
- c. It sets the format to be used for Ordinate Dimensions.
- d. It creates the dimensions automatically without picking points.

5. How would you change a single dimension to a different dimension style, as shown in Figure 26–68?

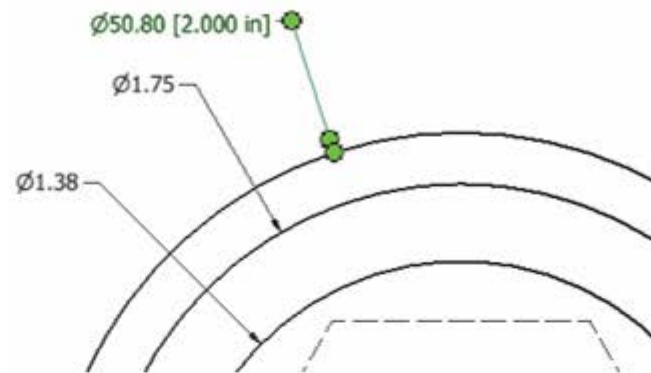


Figure 26–68

- a. Edit the text to change the dimension.
 - b. Delete the dimension and reapply using a different style.
 - c. Select the dimension and choose a different style from the toolbar list.
6. How many sheets can you add to a drawing?
- a. Two
 - b. Odd number
 - c. Even number
 - d. No limit
7. Only a single sheet can be deleted at one time.
- a. True
 - b. False
8. When creating a parts list, how do you set the assembly that is used as the source?
- a. You set the source file in the Styles and Standards.
 - b. The first view created in the drawing is used as the source.
 - c. You set the source file in the Document Settings.
 - d. You select a view in the drawing or browse to a file.
9. When creating a parts list, you can display only top-level components.
- a. True
 - b. False

10. Which of the following statements is true regarding balloons in a drawing view?
- a. Balloons identifying item numbers in a parts list must be individually placed on each component in the assembly view.
 - b. When using **Auto Balloon**, you can use the **Around**, **Horizontal**, or **Vertical** sub-options to align balloons around the view.
 - c. Offset spacing enables you to position balloons at a fixed distance from the model.
 - d. The shape of all balloons driven by the parts list is circular.
11. What controls the appearance of drawing annotations, such as centerlines, default text style, or parts lists?
- a. Document Settings
 - b. Styles Editor
 - c. Sheet Formats
 - d. Application Options
12. What is the process to edit the hatching for a component in a section view, as shown in Figure 26–69?

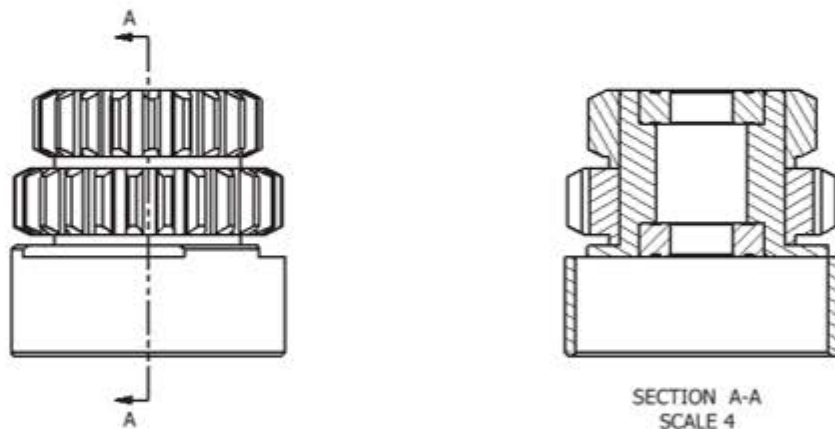


Figure 26–69

- a. In the *Tools* tab, select **Document Settings** and select the *Drawing* tab.
- b. Right-click on the hatch and select **Edit** to open the Edit Hatch Pattern dialog box.
- c. In the *Tools* tab, select **Application Options**, and select the *Drawing* tab.
- d. Delete the section view and create a new one.

Chapter Review Questions

1. You can only create text with a leader.
 - a. True
 - b. False
2. Which of the following are valid symbols that can be created in a drawing? (Select all that apply.)
 - a. 
 - b. 
 - c. 
 - d. 
3. To which of the following features can you add a hole note in a drawing view? (Select all that apply.)
 - a. Holes
 - b. Extruded circular cuts
 - c. Extruded circular solids
 - d. Chamfer
4. Which chamfer note option displays the horizontal distance between a selected edge and a reference line?
 - a.  (Distance 1)
 - b.  (Angle)
 - c.  (Distance 2)

5. Which command is used to automatically place center marks on all holes in a view, as shown in Figure 27–63?

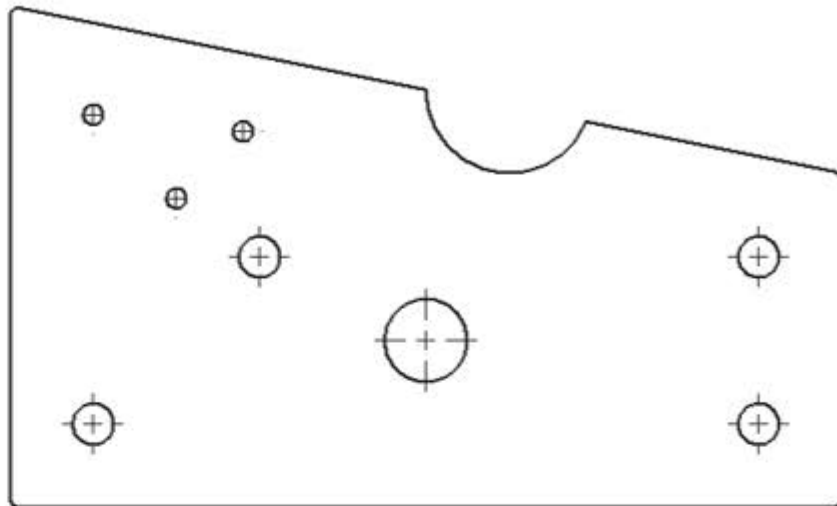


Figure 27–63

- a. **Centerline** tool.
 - b. **Center Mark** tool.
 - c. Right-click on the view and select **Automated Centerlines**.
 - d. Select **All Model Dimensions** in Drawing View dialog box.
6. Fill in the Hole Table type, **Selection**, **View**, or **Features**, with the best description of how it can be used to create a table.
- a. Using the _____ type, all identical holes of all selected types are automatically included.
 - b. Using the _____ type, all holes included in the hole table are selected individually.
 - c. Using the _____ type, all holes of any type in a selected view are included in the hole table.
7. Which of the following drawing annotation types is used to help track and distinguish the various changes in the model between revisions?
- a. Parts list
 - b. Revision table
 - c. Hole table created using **Hole View**
 - d. Balloons